function so we can reinforce that action. The reward signal is also fed into the loss function, which we attempt to minimize with respect to the policy network parameters.

Figure 4.8. Two example environments provided by OpenAI's Gym environment. The OpenAI Gym provides hundreds of environments to test your reinforcement learning algorithms on.

Figure 4.9. A screenshot from the CartPole game environment in OpenAI Gym. There is a cart that can roll left or right, and on top of it is a pole on a pivot. The goal is to balance the pole upright on the cart by carefully moving the cart left or right.

Figure 4.10. After training the policy network to 500 epochs, we get a plot that demonstrates the agent really is learning how to play CartPole. Note that this is a moving average plot with a window of 50 to smooth the plot.

Chapter 5. Tackling more complex problems with actor-critic methods

Figure 5.1. A policy function takes a state and returns a probability distribution over actions, where a higher probability indicates an action that is more likely to result in the highest reward.

Figure 5.2. Q-learning falls under the category of value methods, since we attempt to learn action values, whereas policy gradient methods like REINFORCE directly attempt to learn the best actions to take. We can combine these two techniques into what's called an actor-critic architecture.

Figure 5.3. Read from left to right, raw data is fed into a predict temperature model that predicts the next day's temperature. That prediction is then used in another prediction model that predicts day 2's temperature. We can keep doing this, but initial errors will compound and our predictions will become inaccurate for distant predictions.

Figure 5.4. The bias-variance tradeoff is a fundamental machine learning concept that says any machine learning model will have some degree of systematic deviation from the true data distribution and some degree of variance. You can try to reduce the variance of your model, but it will always come at the cost of increased bias.